STRAVA FITNESS

PowerBI Dashboard Report

Introduction : This document outlines the final phase of the Strava Fitness analysis project which is the creation of an interactive dashboard using Microsoft Power BI. The objective was to connect to the project's data sources and build a user-friendly, visual, and interactive report to present the key findings from the SQL and Python analysis phases. This dashboard serves as the final deliverable for the project.

Data Source Connection : While a direct MySQL connection was initially attempted, it was bypassed due to persistent driver and authentication errors. The final, successful approach was to connect to the raw CSV files stored in the project's folder.

Data Transformation (Power Query) : All data cleaning, which was previously performed in SQL, was replicated in the Power Query Editor to ensure the data model was clean before visualization. Key transformation steps included:

- **Filtering Data**: Loaded only the key tables needed for analysis (e.g., dailyactivity, sleepday, hourlycalories, etc.).

- **Correcting Data Types**: Changed date columns (e.g., ActivityDate) from text to the Date type. The Using Locale feature was used to specify the English (United Kingdom) format(*d/m/y*) to correctly parse the source files.

- **Filtering Rows**: Removed ”non-wear” days by filtering dailyactivity where TotalSteps was not greater than 0.

- **Removing Duplicates**: Replicated the SQL SELECT DISTINCT by selecting the key columns (e.g., Id, SleepDay) and using the Remove Duplicates feature.

- **Replacing Values**: Replicated the SQL SET command for booleans by using Replace

Values to change ”TRUE” to 1 and ”FALSE” to 0 in the weightlog table.

*After these steps, the clean tables were loaded into the Power BI data model by clicking Close & Apply.

Methodology: Data Modeling (DAX) : To enable analysis by day of the week, two calculated columns were added to the dailyactivity table using DAX (Data Analysis Expressions).

3.1 Column: DayOfWeekName

This column extracts the full name of the weekday from the ActivityDate column.

DayOfWeekNameFORMAT('dailyactivity '[ActivityDate], "dddd")

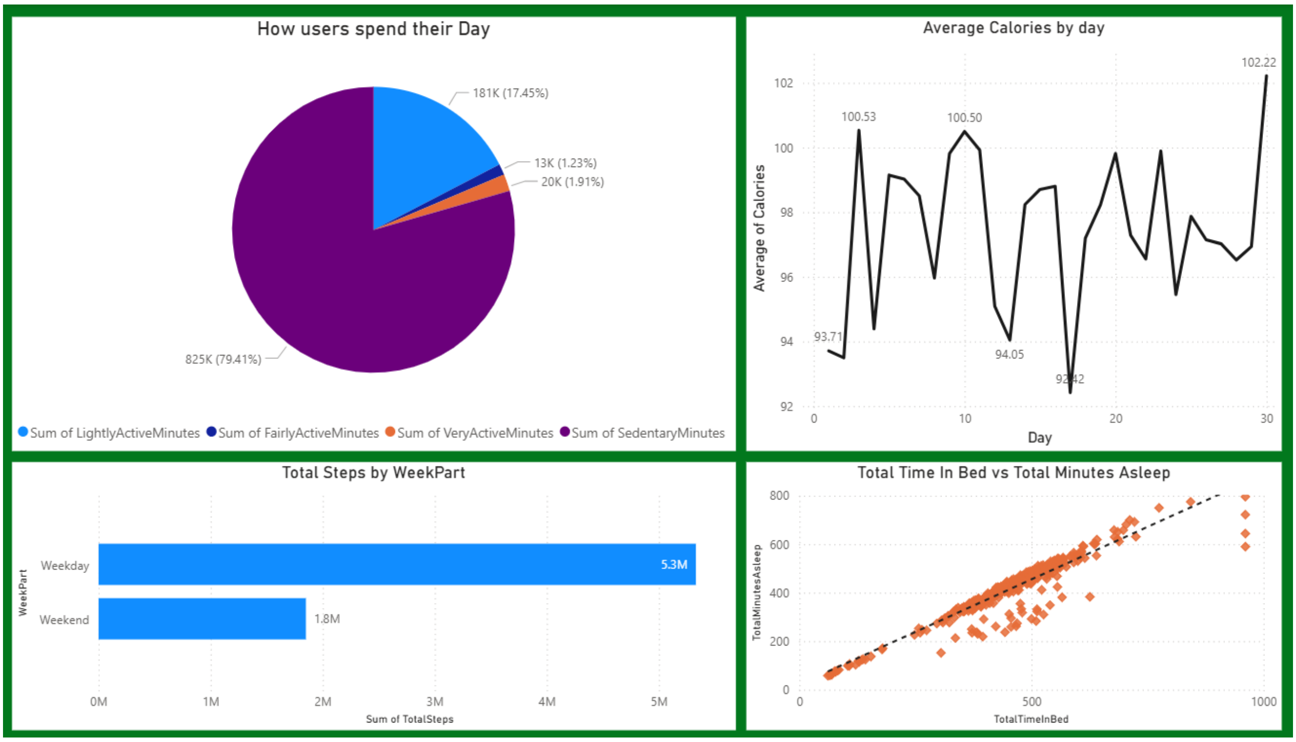
3.2 Column: WeekPart

This column uses the DayOfWeekName to classify each row as either a ”Weekday” or ”Weekend” for aggregation.

WeekPart=IF(

('dailyactivity '[DayOfWeekName] = "Saturday ") | | ('dailyactivity '[DayOfWeekName] = "Sunday "), "Weekend", "Weekday")

Final Dashboard Visualization : The cleaned data and DAX-generated columns were used to build the final interactive dashboard. The dashboard consolidates all key findings from the Python analysis into a single, cohesive view.



Conclusion : This dashboard completes the end-to-end data analysis project. The process involved data extraction, rigorous SQL cleaning, exploratory analysis in Python, and the final assembly of an interactive report in Power BI. The dashboard successfully visualizes user behavior patterns and provides actionable insights from the raw data.

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